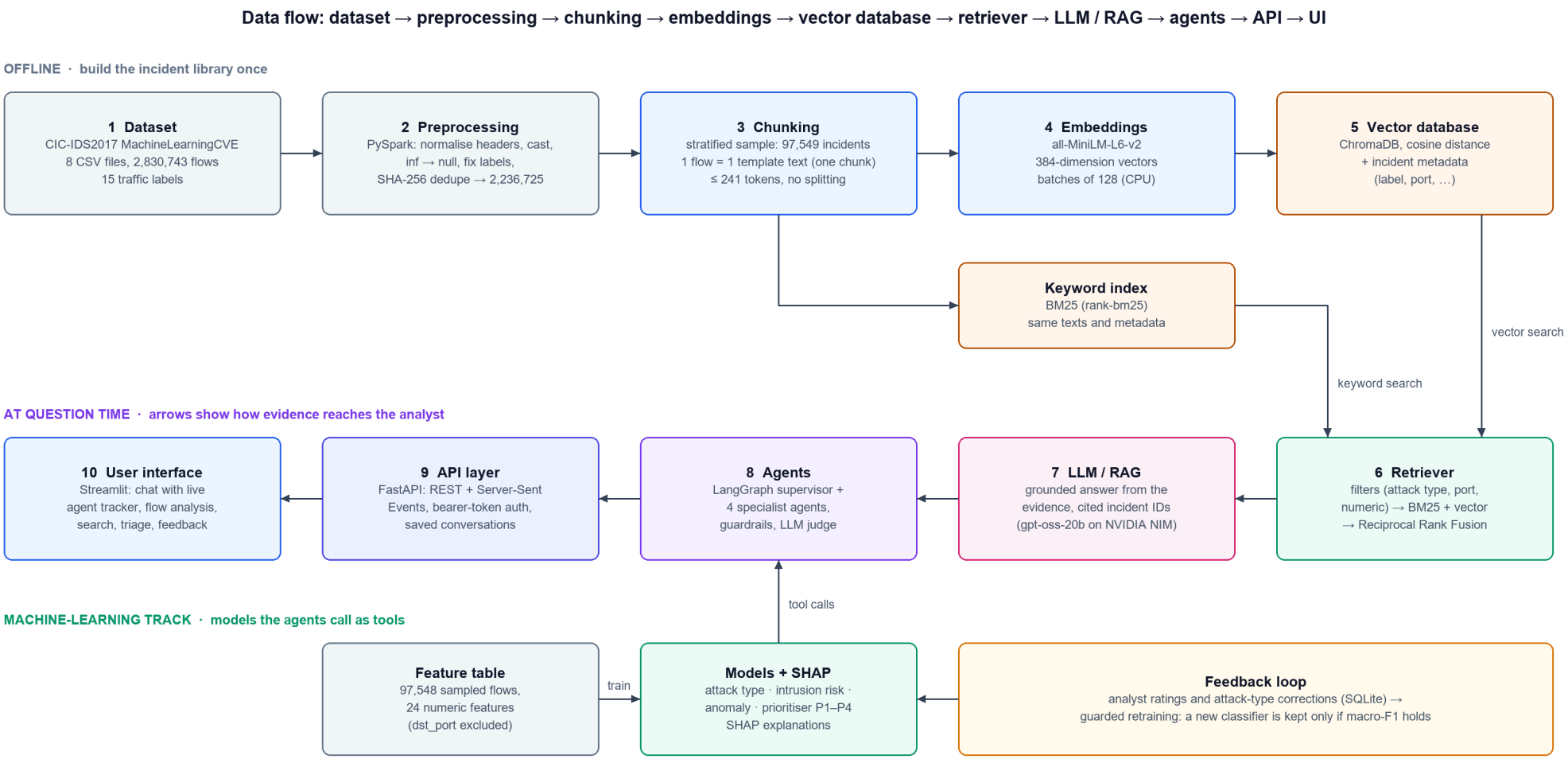
1. Data flow

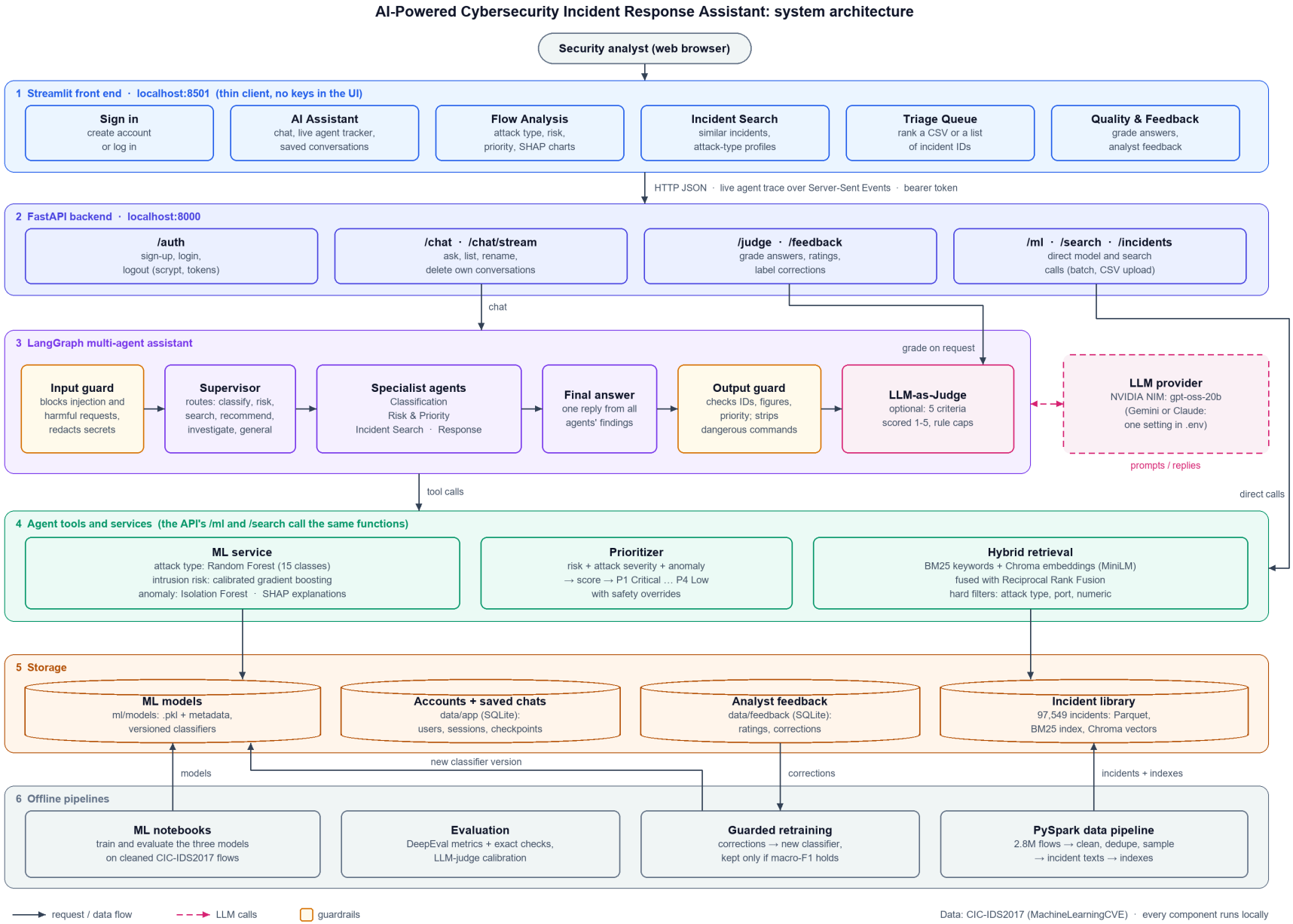


Steps 1–5 run once (about 40 minutes on a CPU). Steps 6–10 run for every question: the question goes from the interface (left) to the retriever (right), and the evidence and answer come back along the arrows.

Each incident is one network flow, so each search hit is a complete, citable incident with its own ID and metadata.

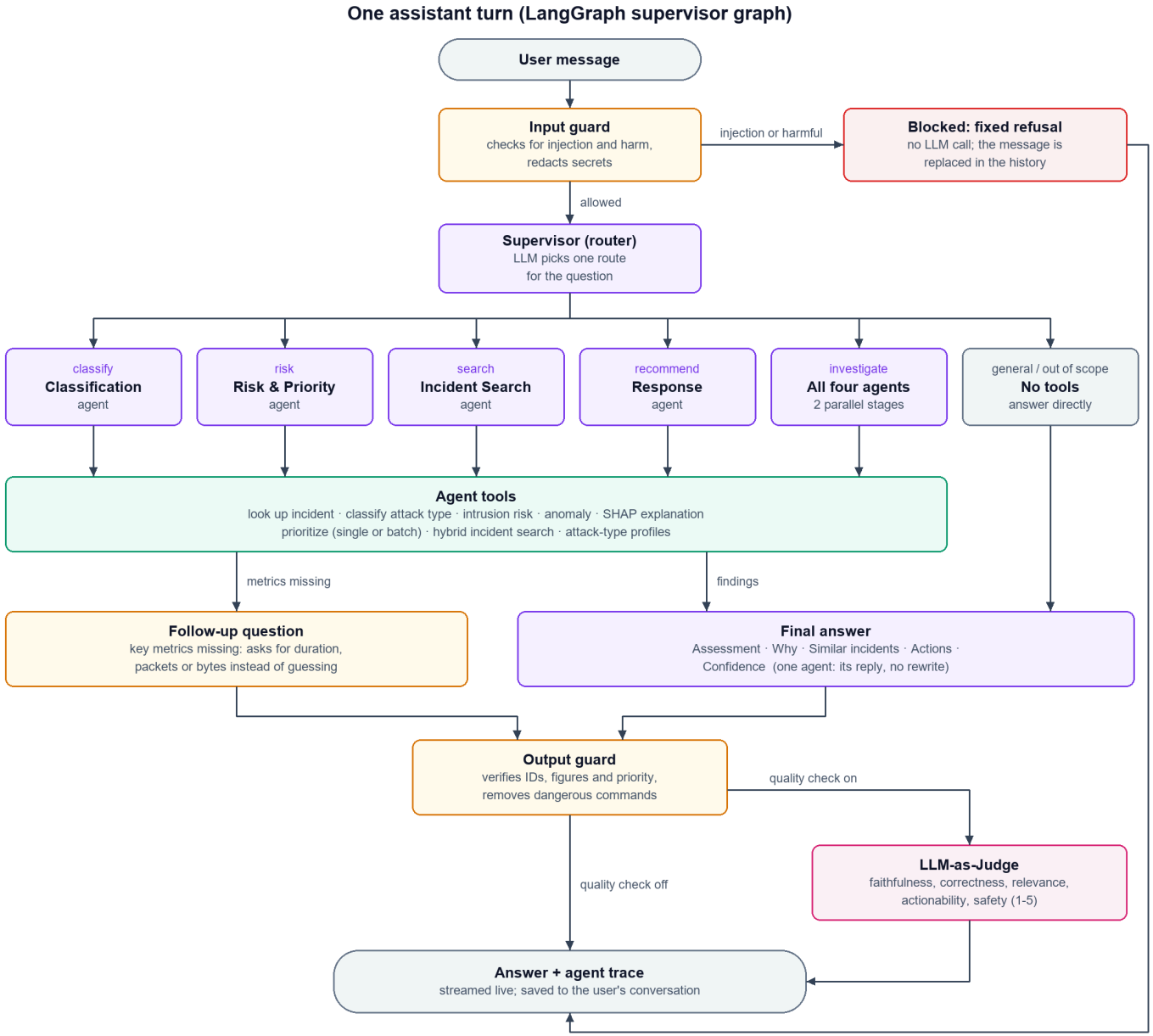
Dataset → preprocessing → chunking → embeddings → vector database → retriever → LLM / RAG → agents → API → user interface, with the machine-learning track and the feedback loop.

2. System architecture



The six layers: Streamlit front end, FastAPI backend, LangGraph multi-agent assistant with the LLM provider, tools and services, storage, and offline pipelines.

3. Agent workflow



One assistant turn: input guard, supervisor routing, specialist agents and tools, follow-up question or final answer, output guard, optional LLM-as-Judge.